PART-A

1. Define IoT.
2. List the Components of IoT.
3. What is meant by MQTT?
4. Write any three applications of AMQP.
5. What is Piconet?
6. What is the importance of IEEE 802.15.4?
7. What are the requirements of wireless sensor networks?
8. What are the key features of M2M?
9. Write a short note on multimedia wireless sensor networks.
10. List the types of industrial automation.

PART-B

11. Explain 6LowPAN's technologies.
12. Explain Constrained Application Protocol (COAP).
13. Describe about MQTT in detail.
14. Explain the working principle of modes of operation of NFC.
15. Explain Bluetooth technology in detail.
16. Explain the node Behaviour in WSN.
17. Explain the types of sensor networks.
18. Explain the role of IoT automation of the healthcare applications in detail by choosing appropriate case study.

ANSWERS

Part-A

1. **IoT (Internet of Things)** refers to a network of physical devices that are connected to the internet and can collect, share, and act on data without needing human involvement.

   Simple Definition

   IoT means everyday objects connected to the internet so they can communicate with each other and perform smart actions.

   Examples

   - **Smart farming sensors** → monitor soil moisture, temperature, water levels.
   - **Smart home devices** → smart lights, smart fans, smart TVs.
   - **Wearables** → fitness bands, smart watches.
   - **Smart cities** → traffic sensors, smart streetlights.

2. **Components of IoT:**

   1. Sensors and Devices
   2. Connectivity (Wi-Fi, Bluetooth, 4G/5G, LoRa, Zigbee)
   3. IoT Cloud or Platform
   4. Data Processing and Analytics
   5. User Interface (Web or Mobile Applications)

3. **MQTT (Message Queuing Telemetry Transport)** is a lightweight messaging protocol used for communication between devices in IoT systems.

   Simple explanation:

   MQTT helps IoT devices send and receive data efficiently using very little internet bandwidth.

   Key points:

   1. It works on a publish–subscribe model.
   2. It is designed for low-power, low-bandwidth devices.
   3. It is commonly used in smart farming, smart homes, sensors, and automation.

4. Write any three applications of AMQP.

1. **Financial Transaction Systems:**
   AMQP is used in banking and stock trading platforms where reliable and secure message delivery is essential.
2. **Enterprise Messaging:**
   Large organizations use AMQP for communication between different applications, services, and business systems.
3. **Cloud and Distributed Systems:**
   AMQP supports message exchange between cloud-based microservices, making it useful in scalable cloud architectures.

5. What is Piconet?

A **piconet** is a small ad-hoc Bluetooth network consisting of one master device and one or more slave devices. It typically supports up to seven active slave devices. Piconets are used for short-range communication such as connecting phones, laptops, headsets, and wireless peripherals. Multiple piconets can interconnect to form a larger network called a scatternet.

6. What is the importance of IEEE 802.15.4?

IEEE **802.15.4** is an important communication standard designed for low-power, low-data-rate wireless networks. It forms the foundation for technologies like Zigbee, WirelessHART, and Thread. Its importance lies in its ability to support long battery life, low-cost devices, secure communication, and reliable networking required for IoT and wireless sensor networks.

7. What are the requirements of wireless sensor networks?

Wireless sensor networks (WSNs) have several key requirements:

1. **Low Power Consumption:** Sensors must operate for long periods on small batteries.
2. **Scalability:** The network must support a large number of sensor nodes.
3. **Reliable Communication:** Data must be delivered accurately even in noisy environments.
4. **Self-organization:** Nodes should automatically form and manage the network without manual setup.
5. **Fault Tolerance:** The system must continue working even if some sensor nodes fail.
6. **Security:** Data needs to be protected from unauthorized access.

8. What are the key features of M2M?

Machine-to-Machine (M2M) communication includes several important features:

1. **Automatic Device Communication:** Devices exchange data without human involvement.
2. **Remote Monitoring and Control:** Machines can be monitored and controlled from remote locations.
3. **Efficient Data Transfer:** M2M uses low-power and optimized communication to transmit data.
4. **Real-time Operation:** It supports real-time alerts, diagnostics, and automated responses.
5. **Scalability:** It can connect thousands of devices in a large system, such as smart grids or smart homes.

9. Write a short note on multimedia wireless sensor networks.

Multimedia Wireless Sensor Networks (MWSNs) consist of sensor nodes capable of capturing and transmitting multimedia data such as images, audio, and video. These networks require higher bandwidth, more processing power, and larger storage compared to traditional sensor networks. MWSNs are used in applications like surveillance, traffic monitoring, disaster management, and smart city systems. They are designed to handle real-time data transmission while maintaining energy efficiency and network reliability.

10. List the types of industrial automation.

Industrial automation is generally classified into four main types:

1. **Fixed Automation:**
   Used for high-volume production with dedicated equipment. Examples: automobile assembly lines, chemical plants.
2. **Programmable Automation:**
   Allows reprogramming for different tasks. Used in batch production systems.
3. **Flexible Automation:**
   Offers quick changeovers between product types, suitable for mid-volume production.
4. **Integrated Automation:**
   Fully automates the entire manufacturing process using robotics, sensors, and computer control systems.

Part-B

11. 6LoWPAN Technology:

6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks) is an important communication technology used in IoT to enable small, battery-powered devices to connect to the Internet using IPv6. Many IoT devices such as temperature sensors, smart meters, and wearables work on low-power wireless networks like IEEE 802.15.4, which cannot directly handle the large size of IPv6 packets. 6LoWPAN solves this problem by providing a lightweight adaptation layer that makes IPv6 suitable for low-power, low-range devices. It allows efficient communication, supports large device networks, and helps IoT systems operate smoothly even with limited energy and memory.

Key Features:

1. Adaptation Layer
   6LoWPAN adds a special adaptation layer between the network and data link layers to support IPv6 on low-power devices.

2. Header Compression
   IPv6 header (40 bytes) is compressed to a few bytes, saving bandwidth and energy for small devices.

3. Fragmentation and Reassembly
   Large IPv6 packets are broken into smaller frames to match the small frame size of IEEE 802.15.4 devices.

4. Mesh Routing Support
   Devices can route data through each other (multi-hop communication), which increases range.

5. Low Power Consumption
   Designed for devices with small batteries, helping them operate for months or years.

6. Supports IPv6 Protocol

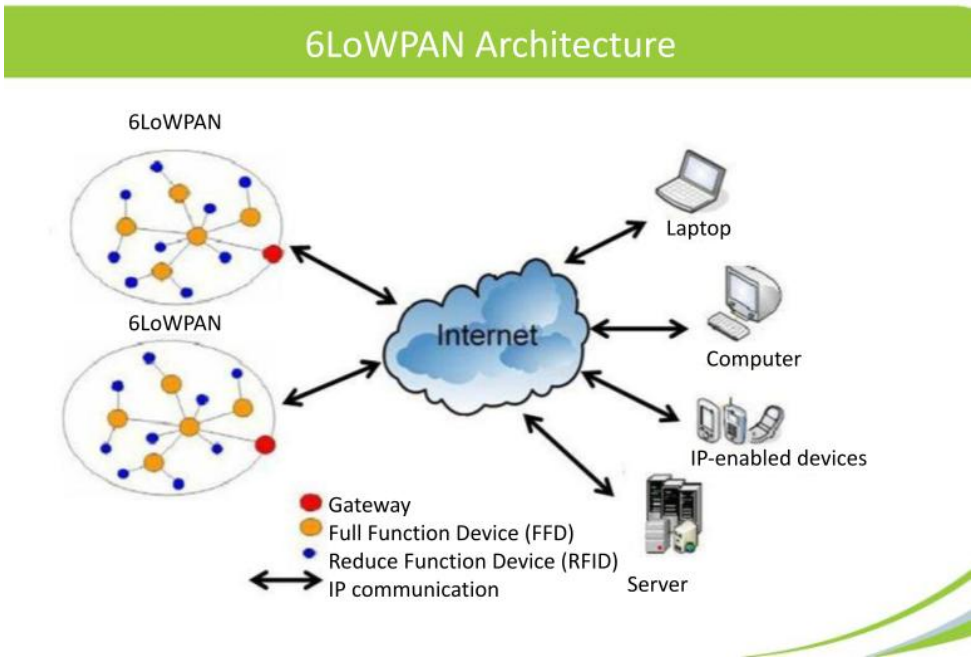
   Makes IoT devices directly connectable to the Internet and cloud services.

7. Interoperability

   Works with many low-power technologies like Zigbee, BLE, and Thread.

8. Scalability

   Suitable for large-scale IoT networks with many devices.



Advantages:

- Energy efficient and suitable for battery-powered IoT devices.
- Supports IPv6 for direct internet connectivity.
- Low-cost and works on simple hardware.
- Provides mesh networking for extended coverage.
- Highly scalable for large IoT networks.
- Uses header compression to save bandwidth.
- Interoperable with Zigbee, BLE, and IEEE 802.15.4.

Disadvantages:

- Low data rate and not suitable for heavy data applications.
- Short communication range without mesh routing.

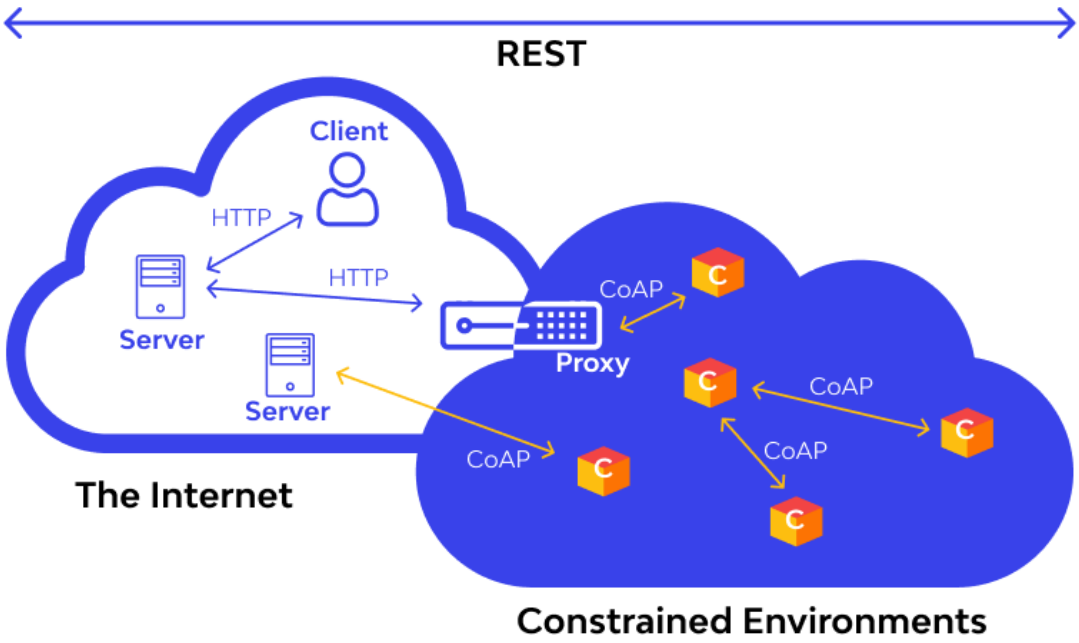
- Limited security unless extra layers are added.
- Setup and routing can be complex.
- Works on devices with low memory and processing power.
- May face wireless interference in crowded environments.

12. COAP:

Constrained Application Protocol (CoAP) is a lightweight communication protocol designed for IoT devices that have low power, low memory, and limited processing capability. It works similar to HTTP, but it is much smaller and faster, making it suitable for tiny devices like sensors, smart meters, and embedded systems. CoAP allows devices to communicate over the internet using the UDP protocol, which is more efficient than TCP for small devices.

CoAP follows a client–server model, where one IoT device (client) sends a request and the other device (server) sends a response. It uses REST architecture, meaning it supports operations like GET, POST, PUT, and DELETE, similar to HTTP. However, CoAP messages are very small and can be transmitted quickly using very little energy. It also supports confirmable and non-confirmable messages: confirmable messages require an acknowledgment, while non-confirmable messages do not.

CoAP is designed for constrained networks, where bandwidth is low and reliability is limited. It supports features like asynchronous communication, multicast, and resource discovery, allowing devices to easily find and communicate with each other. Because CoAP runs over UDP, it is faster, consumes less power, and is suitable for real-time IoT applications like smart homes, healthcare devices, industrial automation, and environmental monitoring.



KeyFeatures:

- Lightweight and designed for low-power IoT devices
- Works on UDP (faster and low overhead)
- Uses REST model (GET, POST, PUT, DELETE)
- Supports confirmable and non-confirmable messages
- Allows multicast communication.
- Supports resource discovery

Advantages:

- Consumes very low power and resources
- Faster communication because it uses UDP
- Easy to implement and suitable for small devices
- Supports REST operations like HTTP
- Allows multicast communication.
- Good for real-time IoT applications.

Disadvantages:

- Not as reliable as TCP-based protocols.
- Security is weaker unless DTLS is added
- May lose packets in unstable networks.
- Not suitable for large data transfers
- Limited support on older network systems.

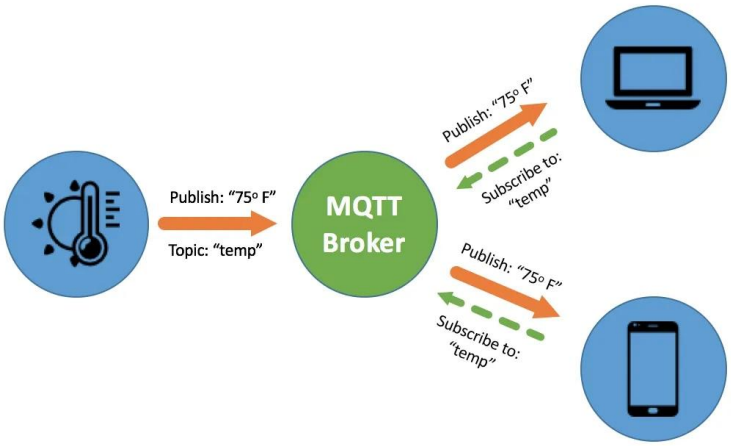
13. MQTT:

MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol widely used in the Internet of Things (IoT). It is designed for devices that have **low power, low memory, limited processing**, and need to send small amounts of data over unreliable or slow networks. MQTT follows a **publish–subscribe communication model**, which makes it highly efficient for IoT applications such as smart homes, wearables, sensors, industry automation, and remote monitoring.

MQTT works using a central system called the **broker**. Instead of devices sending messages directly to each other, they send messages to the broker. Devices that want to **send data** are called *publishers*, and devices that want to **receive data** are called *subscribers*. They communicate using **topics**, which act like channels or labels. For example, a temperature sensor may publish data to the topic "home/room1/temperature," and any device subscribed to this topic will automatically receive the data. This method reduces network traffic and makes the system simple and scalable.

MQTT is built on **TCP/IP**, which ensures reliable message delivery. It supports different **Quality of Service (QoS) levels** to control how messages are delivered:
QoS 0 means message is delivered at most once,
QoS 1 means message is delivered at least once,
QoS 2 means message is delivered exactly once.
These levels help MQTT work efficiently even in unstable network environments.

MQTT is widely used because it is **lightweight**, **fast**, and **energy efficient**. It supports **persistent sessions**, meaning devices that go offline temporarily can receive missed messages when they reconnect. This makes MQTT suitable for remote locations, wireless sensor networks, and mobile IoT devices. MQTT also supports basic security using TCP encryption and authentication, but advanced security needs extra configuration.



Features of MQTT:
• Uses a lightweight publish–subscribe communication model.
• Requires a central broker to manage all messages.
• Works on TCP/IP for reliable data delivery.

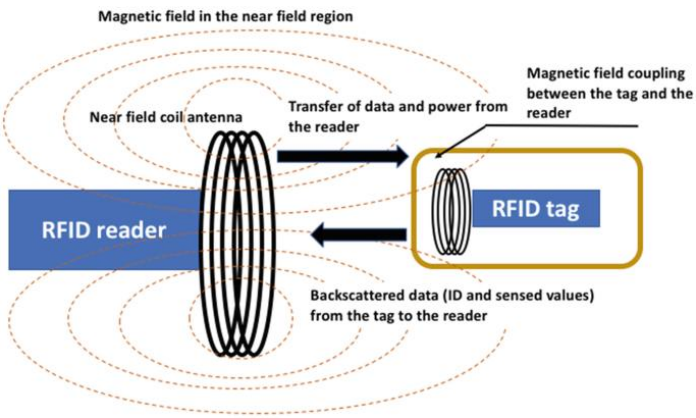
• Supports QoS levels 0, 1, and 2 for message reliability.
• Allows persistent sessions for devices that reconnect.
• Uses topic-based filtering for easy message organization.
• Very low bandwidth and low power consumption.
• Ideal for constrained IoT devices and unstable networks.

Advantages of MQTT:
• Consumes very low bandwidth and power.
• Works efficiently even in weak or unstable networks.
• Publish–subscribe model makes communication simple and scalable.
• Supports reliable delivery using QoS levels.
• Provides persistent sessions for offline devices.
• Easy to integrate with cloud platforms (AWS, Azure, Google Cloud).
• Suitable for thousands of connected IoT devices**.**

Disadvantages of MQTT:
• Requires a broker, increasing setup complexity.
• Security is limited unless strong encryption is added.
• Not suitable for large data transfers like video.
• Based on TCP, which may use more resources than UDP.
• Some legacy devices or networks may not support MQTT.

14. Working Principle of NFC (Near Field Communication):

NFC (Near Field Communication) works on the principle of **electromagnetic induction**, where two devices exchange data when they are brought very close, typically within **4–10 cm**. When an active NFC device creates a magnetic field, it can induce current in another nearby device, enabling communication. NFC operates at **13.56 MHz** and supports low-speed, secure communication. It is mainly used for contactless payments, smart cards, access control, file sharing, and device pairing. NFC communication involves one device acting as the **initiator** and the other as the **target**. The initiator generates the RF field, while the target responds either using its own power (active) or by using energy from the initiator's field (passive). This short-range nature increases security and reduces interference.



Modes of Operation of NFC

1. Reader/Writer Mode:
In this mode, an NFC-enabled device (like a smartphone) reads or writes data to an NFC tag. The phone acts as the initiator, and the tag acts as the passive target. This is used in smart posters, product labels, ID cards, and information sharing.

2. Peer-to-Peer Mode:
In this mode, two NFC-enabled devices communicate directly with each other by exchanging information in both directions. Both devices take turns as initiator and target. This mode is used for sharing files, contacts, photos, and for quick device pairing.

3. Card Emulation Mode:
In this mode, the NFC device behaves like a contactless smart card. It allows smartphones to be used for secure transactions such as mobile payments, metro card access, or digital ID cards. The external reader reads the phone just like a regular NFC card.

Advantages:
• Simple and easy to use with touch-based operation.
• Secure due to very short range.
• Low power consumption, especially in passive devices.
• Can replace cash, cards, and tickets for convenience.
• Supports multiple applications like payments, access control, and device pairing.

Disadvantages:
• Very short communication range (4–10 cm).
• Low data transfer speed, not suitable for large files.
• Can be vulnerable to eavesdropping if additional security is not used.
• Requires devices to have NFC hardware.
• Limited range reduces flexibility in some applications.

15. Bluetooth Technology:

Bluetooth is a short-range wireless communication technology used to exchange data between devices over short distances without using wires. It operates in the 2.4 GHz ISM (Industrial, Scientific, and Medical) radio frequency band and is designed for low-power, low-cost communication between devices like smartphones, laptops, headphones, speakers, smartwatches, IoT devices, and more.

Bluetooth was developed to replace cables for communication between devices and supports both voice and data transmission. It can connect devices within a range of approximately 10 meters to 100 meters, depending on the Bluetooth class. Bluetooth uses frequency hopping spread spectrum (FHSS), which changes frequencies rapidly to reduce interference from other wireless devices in the 2.4 GHz band.

Bluetooth communication works in a master–slave structure. A device acting as a master can connect with up to seven active slave devices in a network called a piconet. Multiple piconets can connect to form a scatternet, allowing larger networks. Bluetooth supports

different profiles that define how devices communicate for specific applications, such as Hands-Free Profile (HFP) for headsets, Advanced Audio Distribution Profile (A2DP) for music streaming, and Human Interface Device Profile (HID) for keyboards and mice.

Advantages of Bluetooth:
• Consumes low power, suitable for portable devices.
• Wireless communication, eliminates the need for cables.
• Supports multiple devices in a network.
• Reliable communication due to frequency hopping.
• Easy to set up and widely compatible.
• Supports both voice and data transmission.
• Secure communication with authentication and encryption.

Disadvantages of Bluetooth:
• Limited range, typically up to 100 meters depending on class.
• Slower data transfer compared to Wi-Fi.
• Interference possible with other 2.4 GHz devices.
• Limited number of active devices in a piconet.
• Not suitable for large file transfers or high-speed applications.

Applications of Bluetooth:

• Wireless headsets and earphones.
• File transfer between smartphones and laptops.
• Smart home devices like lights, locks, and sensors.
• Wearable devices like smartwatches and fitness trackers.
• Wireless keyboards, mice, and game controllers.
• IoT devices and industrial automation.

16. Node Behaviour in WSN (Wireless Sensor Networks):

A **Wireless Sensor Network (WSN)** consists of a large number of sensor nodes that monitor environmental conditions like temperature, pressure, light, or motion. Each node in WSN is **intelligent, small, and energy-constrained**, and its behavior determines the overall performance of the network. The behavior of a node can be described based on its **functional roles, states, and communication patterns**.

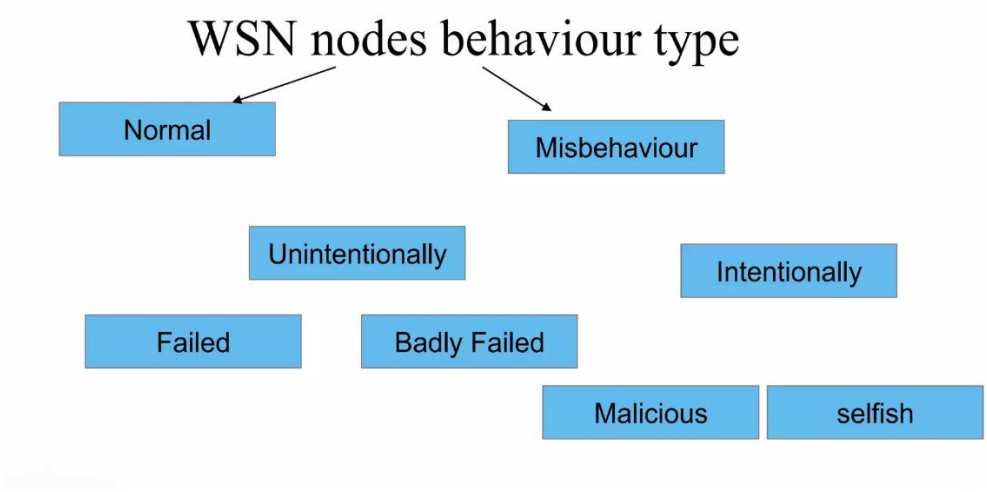
1. Functional Roles of a Node:
Each node can perform several tasks: sensing, data processing, and communication. A node senses physical or environmental data, processes it locally to reduce redundancy, and transmits relevant information to neighboring nodes or a central base station.

2. Node States:
WSN nodes operate in different states to save energy:

- **Active State:** The node is sensing, processing, or transmitting data.
- **Sleep State:** The node shuts down its radio and sensor to conserve energy.
- **Idle State:** The node is ready to receive or transmit but not actively processing data.
- **Dead State:** The node's battery is exhausted, and it can no longer function.

3. Communication Behaviour:
Nodes communicate with each other using **multi-hop routing**, where data is sent through intermediate nodes to reach the base station. They can also collaborate to perform **data aggregation**, reducing network traffic and energy consumption. Nodes dynamically adjust their transmission power, routing paths, and duty cycles depending on network conditions.

4. Adaptive Behaviour:
Nodes may **change roles dynamically** (e.g., from regular node to cluster head in hierarchical routing protocols) to balance energy usage and extend network lifetime. They also monitor their own energy and may reduce activity or delegate tasks to other nodes to conserve power.



Features of Node Behaviour in WSN:
• Nodes can sense, process, and transmit data intelligently.
• Operate in multiple states: active, sleep, idle, and dead.
• Energy-efficient operation to prolong network lifetime.
• Support multi-hop communication for data transmission.
• Can perform data aggregation to reduce network traffic.
• Dynamic role assignment, such as cluster head selection.

• Adaptive to network changes and node failures.
• Collaborate with neighbouring nodes for efficient operation.

17. Types of Sensor Networks (WSN) – Detailed Explanation

A **Wireless Sensor Network (WSN)** is a collection of spatially distributed sensor nodes that monitor physical or environmental conditions such as temperature, pressure, humidity, light, sound, or motion. Each sensor node can sense, process, and communicate data to other nodes or a central base station. WSNs are widely used in environmental monitoring, healthcare, defense, agriculture, smart cities, and industrial automation. Sensor networks are classified based on their deployment environment, mobility, and application requirements.

1. Terrestrial Sensor Networks:
These networks are deployed on land to monitor environmental conditions. Nodes may be scattered randomly, such as in forests for fire detection, or deployed in a planned manner, such as in agriculture to monitor soil moisture and crop health. Terrestrial sensor networks are the most common type of WSN due to their ease of deployment and wide applicability.

2. Underground Sensor Networks:
Used for monitoring underground phenomena like soil moisture, underground water levels, or pipelines. Nodes are buried underground, which poses challenges for signal transmission because radio waves are attenuated by soil. These networks are important in precision agriculture, pipeline monitoring, and geophysical studies.

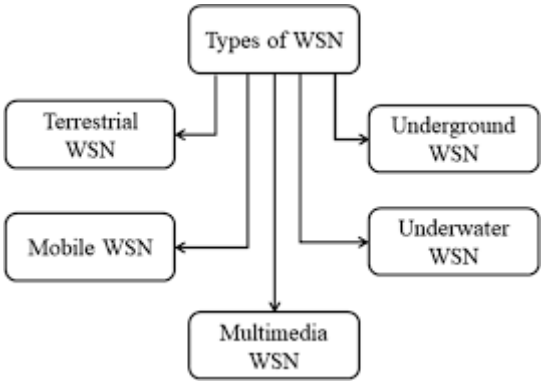
3. Underwater Sensor Networks:
These networks are deployed in water bodies to monitor aquatic environments, including oceans, rivers, and lakes. Nodes use acoustic signals instead of radio frequency signals because water absorbs radio waves. Underwater sensor networks are used for oceanography, pollution monitoring, submarine tracking, and tsunami warning systems.

4. Multimedia Sensor Networks:
Composed of sensor nodes equipped with cameras, microphones, or other multimedia devices. These networks capture audio, video, or image data in addition to traditional sensor measurements. Multimedia sensor networks are used in surveillance systems, traffic monitoring, and smart city applications to provide rich contextual information.

5. Mobile Sensor Networks:
Nodes in these networks are mobile and can move either autonomously or attached to moving objects. Mobile sensor networks are useful in dynamic environments such as battlefield monitoring, wildlife tracking, vehicular networks, or disaster management. Mobility allows these networks to cover larger areas and adapt to changing conditions.

6. Hybrid Sensor Networks:
Hybrid networks combine two or more types of sensor networks to achieve complex monitoring goals. For example, terrestrial nodes can be combined with mobile nodes to

enhance coverage and reliability in environmental monitoring or disaster response systems. Hybrid networks provide flexibility, better scalability, and improved fault tolerance.



Advantages of Sensor Networks:
• Enable real-time monitoring of physical or environmental parameters.
• Reduce human intervention and operational costs.
• Scalable to cover large areas with numerous sensor nodes.
• Useful in diverse applications such as agriculture, healthcare, smart cities, and industrial automation.
• Can detect events or changes quickly and accurately.
• Provide valuable data for decision-making and automation.

Disadvantages of Sensor Networks:
• Limited battery life and energy resources for sensor nodes.
• Communication may be unreliable in harsh or remote environments.
• Deployment in difficult terrains can be challenging.
• Nodes may fail, affecting connectivity and network performance.
• Limited processing and memory capabilities in small sensor nodes.
• Security and privacy are concerns in critical applications.

Features of Sensor Networks:
• Nodes can sense, process, and transmit data intelligently.
• Nodes may be stationary or mobile depending on the type of network.
• Support multi-hop communication to efficiently relay data.
• Energy-efficient operation is critical to prolong network lifetime.
• Can perform data aggregation to reduce network traffic.
• Adaptive to changes in network topology and node failures.
• Highly scalable, suitable for monitoring large areas or complex environments.
• Can operate autonomously with minimal human intervention.

18.Role of IoT in Healthcare Automation

Internet of Things (IoT) has revolutionized healthcare by enabling automation, remote monitoring, and real-time data collection. IoT connects medical devices, sensors, wearable devices, and cloud platforms to collect, process, and analyze patient data. This connectivity allows healthcare providers to monitor patients continuously, detect anomalies early, and provide timely interventions. IoT in healthcare helps reduce human errors, improves patient outcomes, and enhances efficiency in hospitals and clinics.

IoT-enabled healthcare automation includes **remote patient monitoring, smart medical devices, automated alerts, and data-driven decision-making**. Wearable devices like smartwatches and fitness trackers monitor heart rate, blood pressure, glucose levels, and physical activity. Hospital equipment connected to IoT networks can track medication usage, manage inventories, and alert staff about patient emergencies. Automated systems in healthcare reduce manual tasks, optimize workflow, and allow doctors to focus on critical care.

Case Study: Remote Patient Monitoring Using IoT

Scenario: A hospital implements an IoT-based remote patient monitoring system for chronic patients, such as those with heart disease or diabetes. Each patient is provided with wearable devices that continuously monitor vital signs like heart rate, blood pressure, oxygen saturation, and blood glucose.

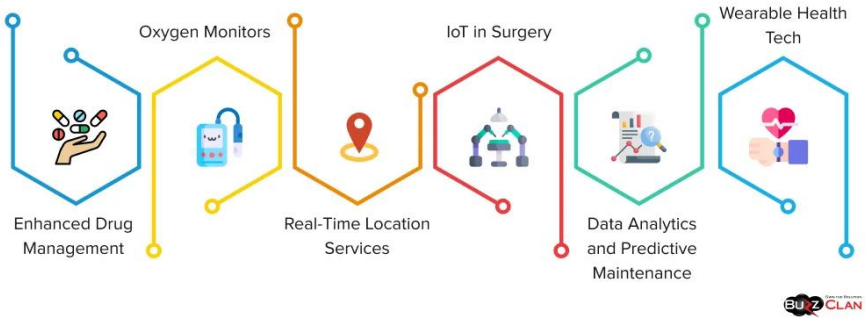
Working:

- The wearable sensors collect real-time data and send it to a cloud-based platform using IoT connectivity.
- The platform analyzes data using AI and machine learning algorithms to detect abnormal patterns.
- If a patient's vitals go beyond safe limits, an **automated alert** is sent to the doctor and the patient via mobile app or SMS.
- Doctors can remotely review patient records, adjust medications, or schedule virtual consultations without requiring physical visits.

Benefits:

- Continuous monitoring improves early detection of critical conditions.
- Reduces hospital visits and healthcare costs.
- Enables personalized care and timely medical interventions.
- Improves patient engagement and adherence to treatment plans.

Example Implementation: Systems like **Philips HealthSuite**, **Medtronic CareLink**, or **Fitbit Health Solutions** are widely used in hospitals and home-care setups. These platforms demonstrate how IoT automation ensures seamless healthcare management.



Key Roles of IoT in Healthcare Automation

- Remote patient monitoring for chronic and critical patients.
- Automated alerts for emergencies and abnormal vital signs.
- Data-driven decision-making using analytics and AI.
- Smart medication management and inventory tracking in hospitals.
- Integration with electronic health records (EHR) for seamless patient care.
- Reducing manual intervention and improving efficiency of healthcare staff.

Advantages of IoT in Healthcare:

- Continuous real-time monitoring of patients.
- Early detection and prevention of critical conditions.
- Reduced hospital visits and associated costs.
- Improved patient safety and adherence to treatment.
- Better resource management and workflow efficiency.

Disadvantages / Challenges:

- Data security and privacy concerns due to sensitive health information.
- High initial cost of IoT-enabled medical devices and infrastructure.
- Dependency on network connectivity and cloud services.
- Complexity in integrating multiple devices and platforms.
- Maintenance and battery life of wearable and sensor devices.

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